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Automobile Insurance Fraud Detection Based on PSO-XGBoost Model and Interpretable Machine Learning Method

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ABSTRACT

Automobile insurance fraud has become a critical concern for the insurance industry, posing significant threats to socio-economic stability and commercial interests. To tackle these challenges, this paper proposes a PSO-XGBoost fraud detection framework and uses explainable artificial intelligence to interpret the predictions. The framework combines an XGBoost classifier with the particle swarm optimization algorithm and is validated through a comparative evaluation against other models. Traditional methods, including SVM, Naive Bayes, Logistic Regression, and BP Neural Network, demonstrate moderate accuracy, ranging from 54.1% to 68.6%, while more advanced models like Random Forest reach up to 78.4%. Compared to the standard XGBoost, the PSO-optimized model achieves 3% superior accuracy, achieving an impressive 95% success rate. Moreover, SHAP is used to extract and visually depict the contribution of each feature to the model's predictions. It turns out that the policyholder's claim amount is the most significant factor in detecting automobile insurance fraud, with other factors such as vehicle type, responsible party, and the insurer's age also considerably influencing the prediction performance. This paper therefore proves that combining the PSO-XGBoost model with SHAP approach can substantially improve the early warning and prevention of automobile insurance fraud.

1. Introduction

Automobile insurance is a cornerstone of the property insurance sector, and plays a pivotal role in the legal framework of many countries. In 2018, the Chairman of the French Insurance Federation reported that insurance corporations expended an astounding 40 billion euros on compensation for damages alone. This expenditure reflected a continuous increase in claim costs, characterized by significant growth in both material and personal injury expenses, averaging a consistent 4% annual growth rate since 2010. As a result, the average cost of motor vehicle insurance rose by 2.9% in 2019. Illustrating with the official data from China, the original insurance compensation expenditure climbed to 1.89 trillion RMB in 2023, marking a substantial year-on-year increase of 21.94%. Specifically, industry exchange data revealed that the automobile insurance sector recorded underwriting profits of 9.035 billion RMB, a decrease of 12.828 billion RMB or 58.67% from the previous year. According to the International Association of Insurance Supervisors (IAIS), an estimated 20% to 30% of global insurance claims are suspected to be fraudulent annually, potentially contributing to the downturn in the insurance sector's underwriting profits.

Fraud is a term with broad and varied implications, as intricately explained by Van Vlasselaer and colleagues. They described fraud as "a rare, deliberate, hard-to-detect, evolving over time, and often well-organized crime that manifests in multiple forms," necessitating the use of advanced analytical tools and methodologies to effectively combat it (Van Vlasselaer et al., 2017). Among the various types of fraud, insurance fraud stands out due to its detrimental impact on the healthy growth of economic order and the challenges it presents for identification and prevention. A 2019 survey by Insurance Europe revealed that insurance fraud—both detected and undetected—accounted for a significant 10% of all expenses in Europe. This is underscored by the fact that fraudulent claims increased to 107,000 in 2018, a 5% rise over the prior year. Insurance fraud can be further classified into automobile insurance fraud, life insurance fraud, and medical insurance fraud. Data from Chinese officials denote motor vehicle insurance as a sector especially prone to fraud, with an overwhelming 80% of total reported

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insurance fraud cases stemming from this industry. Furthermore, automobile insurance fraud constitute 73% of all insurance fraud cases, according to judgments listed on the Judicial Judgment Network, and some of them are still under investigation. This not only results in direct economic losses for companies but also escalates routine management expenses. Consequently, to maintain profitability and offset losses, insurers often raise premiums for honest policyholders. Investigative agencies find it impractical to verify every claim within a specified time frame, and manual detection methods are labor-intensive and error-prone. Therefore, it is crucial to leverage data mining techniques and apply artificial intelligence methods to identify suspicious claims among a vast amount of claim statements.

For fraud detection, where the distinction between legitimate and fraudulent activities is subtle, ensemble learning is well suited for its robust performance over individual models and its superior capability in handling imbalanced datasets. The XGBoost model, an enhanced gradient boosting framework, utilizes tree structures with second-order derivatives and regularization to significantly boost algorithm efficiency. The performance of XGBoost largely hinges on parameter optimization, and manual tuning can be labor-intensive and complex. To address this, Particle Swarm Optimization (PSO) stands out as a strategic method to fine-tune XGBoost parameters, thereby enhancing the prediction accuracy as well as the capacity of insurance companies and investigative agencies to detect fraudulent claims.

To improve the accuracy and the generalization performance of insurance fraud detection models, this paper introduces a novel approach designed to accurately detect and interpret patterns of fraudulent activities in automobile insurance claims. We develop an advanced model based on XGBoost, which is adaptively optimized using PSO, and conduct comparisons with standard XGBoost and other machine learning models to demonstrate the effectiveness of the PSO-enhanced XGBoost in detecting automobile insurance fraud. This hybrid approach could notably expedite the validation process of automobile insurance claims, reducing the time required for legitimacy assessments. Additionally, we apply SHAP (SHapley Additive exPlanations) to increase the interpretability of complex machine learning models in the context of automobile insurance fraud detection, facilitating policymakers to understand the predictions and assess the significance of various features impacting fraudulent claims. By explaining these feature indicators, the paper graphically provides transparency and insight into the factors that influence the model's fraud detection capabilities.

2. Literature review

Given the substantial impact on both financial stability and socioeconomic well-being, the detection of automobile insurance fraud has gained increasing attention from researchers. This field of study generally encompasses two primary research phases: the first phase involves applying statistical and machine learning techniques to analyze insurance claim samples, while the second phase centers on employing various heuristic optimization and ensemble learning algorithms for more advanced fraud detection research.

Early research on automobile insurance fraud detection primarily focused on using statistical and econometric methods to identify key indicators of fraud (Dionne et al., 2013). With advancements in computer technology, algorithms such as Naive Bayes, Decision Tree, and Random Forest have been applied to construct models for detecting automobile insurance fraud. For instance, Nian et al. (2016) proposed a new unsupervised spectral ranking method for anomaly detection (SRA), which utilized the first non-principal eigenvector of a Laplacian matrix for ranking anomalies (Nian et al., 2016). Badriyah et al. (2018) employed k-nearest neighbors and statistical-based methods, demonstrating that the feature selection process significantly improved the detection of fraudulent activities compared to the original procedure without feature selection (Badriyah et al., 2018). By comparing the identification performance of various machine learning algorithms using motor vehicle claims data from a certain insurance company, Itri et al. (2019) found that the Random Forest algorithm outperformed others (Itri et al., 2019). Li et al. (2017) combined different classifiers to propose a multi-classifier system based on Random Forest, Principal Component Analysis (PCA), and Latent Nearest Neighbor methods. They found that the performance depends on the effectiveness and diversity of the weak learners. Employing PCA to transform data at each node into a unique space for optimal splits enhanced the diversity and overall accuracy (Li et al., 2017). Roy and George (2017) as well as Kowsalya and Manoharan (2018) used algorithms such as Decision Tree, Naive Bayes, and Random Forest to create models specifically for detecting automobile insurance fraud, with both studies finding that the Random Forest algorithm exhibited superior performance in their respective datasets (Roy and George, 2017, Kowsalya and Manoharan, 2018).

As artificial intelligence technologies like deep learning advance rapidly, various heuristic optimization and ensemble learning algorithms are now being applied to automobile insurance fraud detection research. Liu (2014) applied data mining techniques with ensemble algorithms such as Stacking, Majority Voting, AdaBoost, and Bagging to improve accuracy in detecting automobile insurance fraud in China (Wei, 2014). Chun et al. (2019) predicted automobile insurance fraud using the NAGA-BP neural network model (Chun et al., 2019). Wang and Xu (2018) highlighted that previous research focused primarily on numerical factors like claim time, while neglecting the analysis and application of text-based information in insurance claims (Wang and Xu, 2018). Jia (2019) used a set of 302 automobile insurance claim records (200 negative and 102 positive cases) for the analysis, and seven indicators were chosen as input parameters for a BP neural network (Lan, 2019). Gupta et al. (2021) proposed a two-stage process for insurance fraud detection, and integrated class imbalance techniques with a TGAN-GBM model (Gupta et al., 2021). Majhi (2021) developed a fuzzy clustering algorithm by combining improved whale optimization algorithm (MWOA) and fuzzy c-means (FCM), and proved its effectiveness with automobile insurance fraud datasets (Majhi, 2021). Meryem (2022) used LIME to explain the predictions of the automobile insurance fraud detection model and ranked the importance of the feature indicators influencing the model's performance (Yankol-Schalck, 2022). Debener et al. (2023) further explored how the supervised and unsupervised learning methods perform in terms of identifying insurance fraud in a large proprietary data set (Debener et al., 2023). Maiano et al. (2023) proposed an automatic system for identifying fraud attempts in the insurance sector and showed that their method outperformed current solutions both quantitatively and qualitatively (Maiano et al., 2023). Nordin et al. (2024) carried out a comparative study and validated that the AdaBoost algorithm can improve the classification performance of the decision tree (Nordin et al., 2024).

To summarize, numerous machine learning-based methods for detecting automobile insurance fraud have been proposed. However, many of these studies have overlooked the critical issue of class imbalance within the datasets and provided limited insights into model interpretability. Based on the above studies and findings, this paper aims to develop a highly effective framework for detecting automobile insurance fraud and offering insights into its predictions. It starts with balancing the class distribution in the automobile insurance claims datasets, followed by utilizing PSO to optimize the relevant XGBoost parameters. Finally, the SHAP method is employed to explain the model's predictions.

3. Methdology

3.1. Model

The current construction of automobile insurance fraud detection model involves the use of several mainstream machine learning algorithms, including Support Vector Machine (SVM), Decision Tree,

Random Forest, Logistic Regression, and Naive Bayes (Kai and Binjie, 2022). For instance, Badriyah et al. (2018) and Li et al. (2019) utilized Logistic Regression, SVM, Decision Tree, and K-Nearest Neighbors to build machine learning models (Badriyah et al., 2018, Li et al., 2019). However, the accuracy of these models in identifying suspicious fraudulent claims is less than ideal. To address this, we employ the XGBoost algorithm, renowned for its robust predictive performance and strong generalization capabilities, as the foundational algorithm for constructing the model. Furthermore, to enhance the model's performance, we integrate the particle swarm optimization algorithm, celebrated for its excellent optimization capabilities.

3.1.1. XGBoost

XGBoost, short for "Extreme Gradient Boosting", is a composite algorithm that combines base learners with weights to achieve excellent data fitting performance. Since its introduction in 2015, XGBoost has gained substantial popularity across data mining, machine learning, and other domains due to its high scalability, robust generalization capabilities, and rapid computation speed (Zhou et al., 2022).

$$\widehat{y}_i = \sum_{k=1}^{K} f_k(x_i),\ f_k \in F \tag{1}$$

Among them, $F = \{f(x) = w_{q(x)}\}(q : R^m \to \{1, 2, \ldots, T\},\ w \in R^T)$ represents a collection of CART decision tree structures, q denotes the tree structure that maps samples to leaf nodes, T stands for the number of leaf nodes, and w indicates the real-valued scores assigned to the leaf nodes. When constructing an XGBoost model, it is essential to search for optimal parameters to minimize the objective function, thereby achieving the goal of building the best possible model. The objective function of the XGBoost model can be divided into the model complexity term Ω and the error term *L*. The objective function can be expressed as:

$$\mathrm{Obj} = \mathrm{L} + \Omega \tag{2}$$

$$L = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 \tag{3}$$

$$\Omega = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{4}$$

Among them, γT represents the $L1$ regularization term, and $\frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$ represents the $L2$ regularization term.

In the process of optimizing and training the model using the training dataset, it is necessary to preserve the original model unchanged and introduce a new function f into the model to minimize the objective function as much as possible. The specific process is as follows:

$$\widehat{y}_i^{(0)} = 0 \tag{5}$$

$$\widehat{y}_i^{(1)} = \widehat{y}_i^{(0)} + f_1(x_i) \tag{6}$$

$$\widehat{y}_i^{(2)} = \widehat{y}_i^{(1)} + f_2(x_i) \tag{7}$$

……

$$\widehat{y}_i^{(t)} = \widehat{y}_i^{(t-1)} + f_t(x_i) \tag{8}$$

Where $\widehat{y}_i^{(t)}$ represents the prediction of the model at the t-th iteration, and $f_t(x_i)$ represents the new function introduced at the t-th iteration. At this point, the objective function can be expressed as:

$$Obj = \sum_{i=1}^{n} \left(y_i - \left(\widehat{y}_i^{(t-1)} + f_i(x_i)\right)\right)^2 + \Omega \tag{9}$$

Perform a second-order Taylor expansion on the objective function. This process is given by:

$$Obj^{(t)} \approx \sum_{i=1}^{n} \left[\left(y_i - \widehat{y}_i^{(t-1)}\right)^2 + 2\left(y_i - \widehat{y}_i^{(t-1)}\right) f_t(x_i) - h_i f_t^2(x_i)\right] + \Omega \tag{10}$$

By removing the constant term, it becomes evident that the objective function is only related to the first and second derivatives of the error function. The objective function can then be expressed as:

$$Obj^{(t)} \approx \sum_{i=1}^{n} \left[g_i w_{q(x_i)} + \frac{1}{2} h_i w_{q(x_i)}^2\right] + \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{11}$$

$$= \sum_{j=1}^{T} \left[\left(\sum_{i \in I_j} g_i\right) w_j + \frac{1}{2}\left(\sum_{i \in I_j} h_i + \lambda\right) w_j^2\right] + \gamma T \tag{12}$$

If the tree structure q is known, the objective function can be used to find the optimal w_j and obtain the optimal value of the objective function. This process can essentially be reduced to solving a quadratic function for its minimum value. Then solutions can be obtained from (13) and (14) as:

$$w_j^* = \frac{-\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \tag{13}$$

$$Obj = -\frac{1}{2} \sum_{j=1}^{T} \frac{\left(\sum_{i \in I_j} g_i\right)^2}{\sum_{i \in I_j} h_i + \lambda} + \gamma T \tag{14}$$

The *Obj* can serve as a scoring function to evaluate the model, where a smaller *Obj*value indicates better model performance. By recursively applying the aforementioned tree construction method, a large number of regression tree structures can be obtained. These structures are then searched using the *Obj* function to find the optimal tree structure. The optimal structures are incorporated into the existing model, resulting in the establishment of an optimal XGBoost model.

3.1.2. PSO algorithm

The PSO algorithm draws inspiration from the social foraging behavior of bird flocks. Based on their intuitive understanding of the algorithm, they abstracted birds during foraging as particles in space, with each particle corresponding to a solution (Jiang et al., 2023). The process of bird migration can be likened to the continuous optimization of the solution swarm, where finding food signifies that the corresponding particle swarm has located the global optimum.

At the initial stage of the algorithm, a group of particles and their velocities are randomly initialized. The number of particles in the group, referred to as the population size, can be represented by m. Each particle has a position in an n-dimensional space, denoted as $x_i = (x_{i1}, x_{i2} \ldots, x_{in})$, where i ranges from 1 to m. Each position corresponds to a solution to the problem, and particles search for new solutions by continuously altering their positions, x_{in}. Additionally, each particle has a velocity, $v_i = (v_{i1}, v_{i2}, \ldots, v_{in})$, which determines the displacement of the particle in the search space as iterations proceed. The fitness value of each particle is calculated, typically based on the requirements of the actual research problem. In this paper, the fitness function is defined as the error between actual and predicted results. Based on the fitness of each particle, the individual best solution (pbest) found by each particle and the overall best solution (gbest) encountered by the entire particle swarm are updated. Each particle dynamically updates its velocity and position by tracking its individual best value and the global best value using Equation (15) and Equation (16).

$$v_{ij}(t+1) = v_{ij}(t) + c_1 r_1 \left[p_j(t) - x_{ij}(t)\right] + c_2 r_2 \left[g_j(t) - x_{ij}(t)\right] \tag{15}$$

$$x_{ij}(t+1) = x_{ij}(t) + v_{ij}(t+1) \tag{16}$$

Among them, j=1, 2, ⋯, n; t represents the number of iterations; $r1$ and $r2$ are random numbers uniformly distributed between 0 and 1; $c1$

and $c2$ are acceleration factors.

To avoid particles moving too fast and potentially overshooting the optimal solution, an upper limit, v_{max}, is typically set on the particle's velocity. The value of v_{max} can significantly impact the search behavior of the particle swarm: a larger v_{max} enhances the global search capability, while a smaller v_{max} strengthens the local search capability. If v_{max} is set too high, particles might "fly" past the optimal solution, whereas too low a v_{max} may cause particles to remain in the vicinity of their current positions, leading to the search results being trapped in local optima.

To balance the global and local search capabilities, an inertia weight, denoted as "w", is introduced to the particle's flight speed. This can be described as:

$$v_{ij}(t+1) = wv_{ij}(t) + c_1 r_1 \left[p_j(t) - x_{ij}(t)\right] + c_2 r_2 \left[g_j(t) - x_{ij}(t)\right] \tag{17}$$

$$x_{ij}(t+1) = x_{ij}(t) + v_{ij}(t+1) \tag{18}$$

In general, the inertia weight w is typically chosen as within the range of [0, 1]. As w approaches 1, the PSO algorithm exhibits stronger global search capability. Conversely, as w approaches 0, the algorithm's local search capability becomes stronger.

The main steps of the PSO algorithm are as follows:

1. Initialize the particle swarm. Randomly initialize the positions and velocities of the particles based on relevant parameters such as population size.
2. Calculate the fitness value of each particle. Determine the fitness function based on the specific problem and obtain the fitness value for each particle.
3. Update the individual and global best values. Compare the current fitness value with the individual best and global best values. If the current fitness value is better, update the individual and global best values accordingly.
4. Update the positions and velocities of the particles continuously.
5. Check termination condition. If the condition is met, end the PSO algorithm and obtain the best result. Otherwise, return to step 2 and repeat steps 3, 4, and 5 until the termination condition is satisfied. The specific process is shown in Fig. 1.

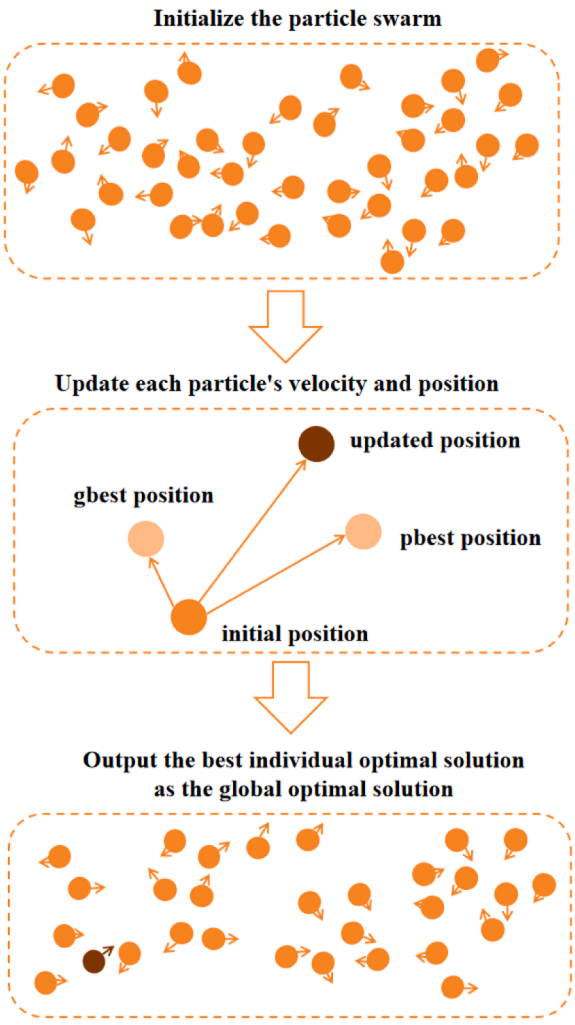


Fig. 1. The iterative process of PSO algorithm

3.1.3. PSO-XGBoost model

The PSO-XGBoost model for automobile insurance fraud detection is designed to enhance the efficiency of public security agencies in investigating and combating automobile insurance fraud. The process begins with data preprocessing on real motor vehicle insurance claims, which includes cleaning, format conversion, and deduplication. This is followed by feature engineering, where statistical features and transformation techniques are applied to prepare the data for computational processing. The model then utilizes PSO algorithm and XGBoost to build the fraud detection framework. The PSO algorithm optimizes the parameters of XGBoost model, improving its ability to identify fraudulent cases. Finally, the model's performance is evaluated using metrics such as accuracy, recall rate, precision, type II error rate, and F1 score. The technical roadmap for constructing the automobile insurance fraud detection model is illustrated in Fig. 2.

The identification of automobile insurance fraud is a typical binary classification problem, where the fraud detection model needs to determine whether a given automobile insurance claim exhibits fraudulent attributes or not. Traditional methods employ decision tree, naive Bayes, SVM, logistic regression, and random forests for classification after data processing. However, these approaches often suffer from issues such as low model accuracy and inadequate fraud detection performance. To develop a more efficient and accurate model for detecting automobile insurance fraud, this paper employs a fusion of XGBoost and PSO algorithms to enhance the effectiveness of the automobile insurance fraud detection model.

XGBoost aims to minimize a regularization objective function and uses decision trees as the base learners to construct the classification model. Therefore, the performance of the model is heavily determined by hyperparameters related to the tree structure (Xie et al., 2022). The number of base learners plays a pivotal role: increasing this number enhances the model's fitting ability but also raises the risk of overfitting and computational time, whereas reducing it diminishes the model's learning capacity, leading to poor recognition performance. The maximum depth of the trees is a critical parameter. In claim datasets with numerous samples and features, increasing the tree depth might be suitable to better capture data complexity (Chen and Guestrin, 2016). The tree feature sampling rate in XGBoost controls the usage of all features in each tree, which helps to prevent overfitting. During tree construction, XGBoost learns leaf weights and employs regularization techniques, such as penalizing the number of terminal nodes and leaf weights, to effectively manage the model's complexity (Nielsen, 2016). The learning rate is another important parameter that affects model performance, whose value directly impacts the learning efficiency of the model (Zhou et al., 2021). Therefore, this paper optimizes the above-mentioned four hyperparameters that significantly influence the model's performance.

The optimization of model hyperparameters is fundamentally a black-box function optimization problem. In practice, having numerous hyperparameters to optimize can result in model redundancy and increased computational complexity, ultimately impacting the overall performance of the model (Tiantian, 2019). XGBoost has a large number

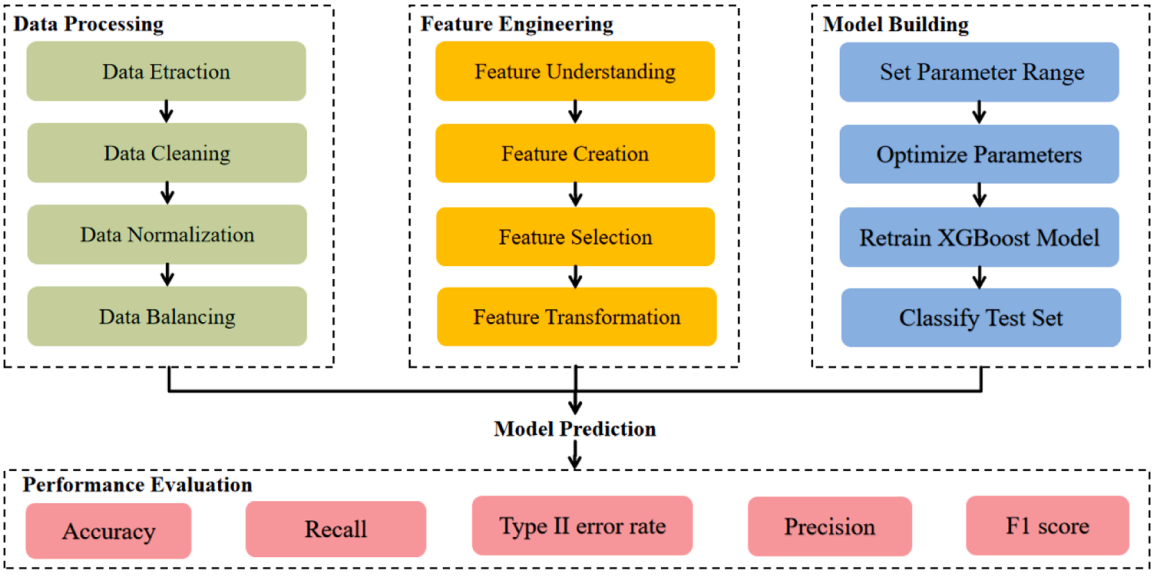


Fig. 2. Roadmap for building a automobile insurance fraud detection model

of parameters, and inaccurate settings can result in lower prediction accuracy. Therefore, determining the relevant parameters of the model requires a trade-off between recognition accuracy and complexity (Zhizhong, 2022). There are two commonly used methods for determining the values of hyperparameters. The first method relies on manual experience to tune the parameters, which often requires a large number of experiments and extensive domain knowledge. The second method involves using heuristic optimization algorithms to tune the parameters, which not only has high efficiency but also allows for obtaining the optimal parameter combination (Pan et al., 2022, Ye and Schuller, 2021). Therefore, in this paper, the PSO algorithm is selected as it has fewer parameters, faster speed, and excellent optimization capabilities to optimize the relevant parameters of XGBoost.

3.1.4. Model evaluation

The confusion matrix is an important evaluation criterion for binary classification models and a vital component of model evaluation. It includes four measurements: true positive (*TP*), false positive (*FP*), true negative (*TN*), and false negative (*FN*), which are illustrated in Table 1 below. True positive (*TP*) refers to instances that are actually positive and correctly predicted as positive by the model. False positive (*FP*) denotes cases that are actually negative but mistakenly identified as positive by the model. True negative (*TN*) represents samples that are genuinely negative and accurately predicted as negative. False negative (*FN*) describes instances that are truly positive but incorrectly classified as negative by the model.

Based on the aforementioned confusion matrix, the following commonly used evaluation metrics can be defined: *accuracy, recall, precision, type II error rate, and F1 score* (An, 2018). Accuracy refers to the proportion of correctly predicted samples out of the total number of samples. The specific calculation formula is as follows:

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN} \tag{19}$$

Table 1
Confusion matrix

Predicted Class	True Class	
	1	0
1	*TP*	*FP*
0	*FN*	*TN*

The recall, also known as sensitivity, measures the proportion of actual positive samples that are correctly predicted as positive by the model.

$$Recall = \frac{TP}{TP + FN} \tag{20}$$

Precision, or positive predictive value, indicates the proportion of samples predicted as positive that are indeed positive.

$$Precision = \frac{TP}{TP + FP} \tag{21}$$

The Type II error rate, also referred to as the false negative rate, represents the proportion of actual positive samples that are incorrectly predicted as negative by the model. This is calculated out of all the actual positive samples in the training or prediction set.

$$FNR = \frac{FN}{FN + TP} \tag{22}$$

The F1 score, which is the harmonic mean of precision and recall, ranges between 0 and 1, with higher values indicating better model performance.

$$F1score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{23}$$

3.2. Interpretable machine learning: SHAP

The SHAP method is an abbreviation for SHapley Additive exPlanations, which combines Shapley's idea with the LIME theory to create an additive explanation model (Lundberg and Lee, 2017). The goal of SHAP is to explain individual predictions by computing the contribution of each feature to the prediction. For each prediction sample, the model generates a predicted value, and the SHAP value represents the allocation of each feature in that sample. This relationship can be expressed as:

$$g(z') = \varnothing_0 + \sum_{j=1}^{M} \varnothing_j z_j' \tag{24}$$

Where g is the explanatory model, M is the total number of input features, $\varnothing_0$ represents the mean prediction. $\varnothing_j \in \mathbb{R}$ is the SHAP value of feature j, which represents the feature attribution. $z' \in \{0,1\}$ is an indicator vector. For a particular sample, if feature *j* is not in the decision path, z' is 0, and the corresponding $\varnothing_j$ is also 0, indicating that feature *j*

has no contribution to the final prediction. Since structured data is used as input in ensemble tree models, for a given sample x, all feature values are present. Therefore, the formula can be simplified as follows:

$$g(z') = \varnothing_0 + \sum_{j=1}^{M} \varnothing_j \tag{25}$$

The SHAP value for each feature j is obtained by summing up the weighted contributions of that feature to the prediction across all possible feature combinations:

$$\varnothing_j(val) = \sum_{S \subseteq \{x_1, x_2, \ldots x_p\} \setminus \{x_j\}} \frac{|S|!(p-|S|-1)!}{p!} (val(S \cup \{x_j\} - val(S)) \tag{26}$$

In the equation, S represents a subset of features used in the model, x is the feature value vector of the sample to be explained, p denotes the number of features, $val(S)$ refers to the predicted value under the feature combination S. The term $\sum_{S \subseteq \{x_1, x_2, \ldots x_p\} \setminus \{x_j\}} \frac{|S|!(p-|S|-1)!}{p!}$ represents the weight. Considering the order, there are $p!$ combinations for p features. Fixing a specific feature j, there remain $(p-|S|-1)!S!$ combinations. $(val(S \cup \{x_j\} - val(S))$ indicates the contribution change before and after adding feature i. By comparing the attribute values of each feature, we can rank the importance of the features.

Due to the large number of subsets S itself, calculating SHAP values using this method can be time-consuming. However, TreeSHAP, a variant of SHAP specifically designed for tree-based machine learning models, offers fast computation speed. It can accurately calculate SHAP values and provide reliable estimates even when there are correlations among features (Lundberg et al., 2018). TreeSHAP is suitable for tree-based models such as Random Forest, XGBoost, GBDT, among others.

4. Data and results

4.1. Data processing

4.1.1. Data

The data used in this paper comprises three parts: investigative agency data on automobile insurance fraud, data from insurance institutions, and international data sources from the Kaggle website (www.kaggle.com). In data cleaning, bad data will be removed, and missing data will be added. To maintain the diversity and variability of the dataset, this paper adopts the method of filling missing values with the mean of the sequence by using IBM's SPSS Statistics 25.0 software. Finally, a total of 11,565 records are obtained.

Each record in the dataset has well-defined attributes and is categorized as either fraudulent claims labeled as "1" or non-fraudulent claims labeled as "0". The dataset comprises a total of 11,565 records, of which only 685 are fraudulent claims, while 10,880 are normal accident claims, indicating a high degree of imbalance. the dataset used in this research presents a significant imbalance, with the ratio of fraudulent claims to legitimate claims being approximately 1:16. This indicates that fraudulent claims are a minority class compared to legitimate claims. Provost and Weiss (2011) suggest that predictive models built on imbalanced datasets tend to minimize the overall error rate in predictions, often masking the high prediction error rate for the minority class (in this paper, the minority class represents "fraudulent claims"). In other words, the model tends to favor the majority class. However, in practice, the high prediction error rate for the minority class can have more severe consequences (Provost and Weiss, 2011). This means that the model may fail to accurately identify fraud, resulting in an inability to effectively combat this type of crime, thereby affecting the effectiveness of crime control. On the other hand, if the majority class has a high error rate, it means that the model predicts normal claims as fraudulent claims, leading to wasted investigation resources and increased costs in terms of personnel and resources (Ding et al., 2021). Therefore, it is necessary to consider both types of error rates comprehensively.

To tackle the challenge of data imbalance, strategies are implemented at both the data and algorithmic levels. At the data level, resampling techniques are employed, which involve adjusting the proportions of the majority and minority classes to create a more balanced dataset. At the algorithm level, enhancements are made by tuning the parameters of the classification algorithm to improve its performance on minority class samples (Chen, 2018). Considering the practicality of modeling data and ease of implementation, this paper utilizes a data-level approach. Specifically, random under sampling is applied to the majority class samples, while random oversampling is performed on the minority class samples. This in turn helps he predictive model to get less biased towards the majority class samples and to achieve better classification performance.

XGBoost inherently excels at handling imbalanced data by increasing the weight of positive samples (fraudulent claims), which allows the model to better identify fraudulent cases. Additionally, the objective function includes $L1$ and $L2$ regularization terms, which help control the model's complexity and prevent overfitting. When dealing with imbalanced data, tuning the strength of these regularization terms can reduce overfitting to the majority class (legitimate claims) while enhancing the detection ability for the minority class (fraudulent claims). Specifically, $L1$ regularization introduces sparsity, simplifying the model structure; whereas $L2$ regularization suppresses excessively large weights, thus balancing the model's sensitivity to different classes.

4.1.2. Feature engineering

The selection of indicator features for each record is informed by insurance industry recommendations, expert insights, literature summaries, and anti-fraud regulations. These factors are integrated to form the indicator features. Each data record includes the following indicator features, as shown in Table 2.

Value assignment for non-numeric variables is essential in feature indicators, involving categorical variables such as "fraudulent status", "gender", "accident area" and "marital status". In the process of model construction, it is necessary to convert categorical variables into numerical variables that can be recognized by computers. For instance, the gender variable, which includes attributes like male and female, can be encoded as 0 and 1, respectively. Likewise, for the marital status variable, which includes attributes like single, married, divorced, and widowed, they can be assigned values of 0, 1, 2, and 3, respectively. This encoding extends to other similar variables.

The goal of feature transformation is to scale the variables to a specific ratio to eliminate disparities in scale among different feature indicators. The dataset contains both numeric and categorical variables with distinct value ranges. Due to significant differences between these variable types, data normalization is necessary. Given that the dataset used in this article comprises tens of thousands of records, with some feature variables being continuous numerical variables with unknown maximum and minimum values, Z-score normalization is adopted. Z-score normalization transforms raw data by adjusting it to have a mean of 0 and a standard deviation of 1. The transformed variables follow a standard normal distribution, and the specific calculation formula is as

Table 2
Names of data feature variables

Number	Names of Feature Variables	Number	Names of Feature Variables
1	FraudFound_P	9	VehiclePrice
2	Sex	10	Deductible
3	MaritalStatus	11	PastNumberOfClaims
4	Age	12	AgeOfVehicle
5	AccidentArea	13	PoliceReportFiled
6	Make	14	WitnessPresent
7	Fault	15	NumberOfCars
8	VehicleCategory	16	ClaimSize

follows:

$$x_{jk}^{*} = \frac{x_{jk} - \mu}{\sigma} \tag{27}$$

In the equation, x_{jk} represents the value of the kth feature variable in the sample x_j, μ represents the mean of the kth feature variable in the original dataset, σ represents the standard deviation of the kth feature variable, and x_{jk}^{*} represents the standardized value of the kth feature variable in the jth sample x_j after normalization. This data processing procedure was performed using SPSS Statistics 25.0 software, and the standardized feature variables are shown in Table 3 as follows:

4.2. Parameter optimization

The algorithms and models used in this paper were obtained from the SPSSPRO data mining software. To evaluate the performance of the automobile insurance fraud detection model, the dataset of vehicle insurance claims used in this paper was divided into a training set and a test set. The training set comprised 80% of the total samples, while the test set contained 20% of the total samples. The training set was used for constructing the fraud detection model, while the test set was used for evaluating the model's performance.

The present study developed the PSO-XGBoost model for identifying fraudulent vehicle insurance claims. Firstly, the PSO parameters were configured. Next, the PSO algorithm was applied to search for optimal parameters for XGBoost. Finally, suitable parameters were obtained and set. In the initial steps, the particle population was set to 50. Selecting 50 particles as the initial value, each representing a potential solution, ensures good coverage. By setting up a loop from 1 to 50, generating a random solution that fits the problem's search space during each iteration, and initializing it as a particle, we can achieve better initial exploration. Using 50 particles allows for covering a wider search space in the early stages, which helps in finding better solutions while keeping computational costs manageable. The output is generated when the number of iterations reaches 100, and the non-negative inertia weight w was set between 0 and 1, with a value of 0.9. These 100 iterations provide ample time for the particle swarm to converge to an optimal solution without excessively long runtime, thus maintaining good time efficiency. By setting an inertia weight of 0.9 for each velocity update, the algorithm can avoid getting stuck in local optima during the early global search phase. Gradually reducing this weight promotes more localized and refined searches, thereby improving model accuracy. Maintaining a higher inertia weight at the beginning encourages broad exploration of the parameter space, while the gradual reduction shifts the focus towards finer local searches as iterations progress. Additionally, the individual learning factor and social learning factor are crucial for model training. These factors respectively affect the particle's learning from its own historical best position and the global best position. Setting both factors to 2 provides balanced individual and social learning capabilities, enabling particles to maintain diversity while effectively utilizing global experience. This ensures the algorithm can balance between global and local search effectively. By configuring these parameters, the PSO-XGBoost model for automobile insurance fraud detection can achieve comprehensive exploration of the search space, enhance convergence speed, and improve the quality of solutions, ultimately resulting in more optimal fraud detection outcomes. The search range for XGBoost algorithm-related parameters is shown in Table 4 below:

4.3. Results and comparison

Utilizing the PSO algorithm to explore the specified parameter ranges, it was discovered that the model exhibits optimal performance with the following settings: 200 base learners (trees), a maximum tree depth of 10, a learning rate of 0.99772, and a tree feature sampling rate of 0.98922. These parameters were then used to run the model with five-fold cross-validation, and the results are summarized in Table 5. Comparing the PSO-XGBoost fraud detection model with the XGBoost model, it can be observed that the PSO-XGBoost model achieved an accuracy of 95%, marking a 3% improvement. Additionally, it showed a 2.4% increase in precision, a higher F1 score, and a lower Type II error rate. In the context of automobile insurance fraud detection, even small improvements in performance can enable the system to identify more difficult-to-detect fraudulent claims, thereby allowing for more effective preventive measures. In the field of automobile insurance fraud detection, adopting the PSO-XGBoost model has significantly reduced the Type II error rate from 8.0% to 5.0%, representing a 37.5% decrease in missed identifications. This improvement underscores the model's enhanced robustness and stability achieved through particle swarm optimization, with statistical significance confirming that this enhancement is a genuine improvement rather than merely due to data fluctuations. It performs more consistently across different datasets. In practical terms, reducing missed identifications enables more effective detection of potential fraud, helping insurance companies minimize losses and enhance the quality of decision-making. Furthermore, more accurate fraud detection improves the user experience by ensuring that legitimate customers receive faster and fairer claims processing.

In order to verify the performance of the PSO-XGBoost model in handling complex and high-dimensional automobile insurance claim data, this paper conducted a comparative analysis with commonly used classification algorithms, including Naive Bayes, Random Forest,

Table 4
XGBoost hyperparameter search range

Parameter Name	Parameter Search Range
Number of Base Learners	[5-200]
Maximum Tree Depth	[1-10]
Learning Rate	[0.1-1]
Tree Feature Sampling Rate	[0.1-1]

Table 3
Standardized feature variables

Number	FraudFound_P	Sex	MaritalStatus	Age	……	ClaimSize
1	0	0	1.47584	-1.62139	……	1.20658
2	0	1	1.47584	-0.55644	……	1.34624
3	0	1	-0.63472	0.50851	……	1.80809
4	0	1	-0.63472	1.98306	……	-0.57011
5	0	1	-0.63472	-0.3926	……	2.80085
6	0	1	1.47584	-0.88411	……	1.26739
7	0	1	1.47584	2.47457	……	1.46964
8	0	1	-0.63472	0.91811	……	2.98153
9	0	1	-0.63472	-1.04795	……	1.36303
10	0	1	1.47584	0.00061	……	2.19554
……	……	……	……	……	……	……
11565	1	1	1.47584	-1.62139	……	-0.79168

Table 5
Comparison between XGBoost and PSO-XGBoost

Model	Accuracy	Recall	Precision	Type II error rate	F1
XGBoost	92%	92%	93%	8.0%	0.921
PSO-XGBoost	95%	95%	95.4%	5.0%	0.950

Logistic Regression, Decision Tree, SVM, and BP Neural Network, as well as the XGBoost classification model used in constructing the PSO-XGBoost model, under the same experimental environment. From Table 6, we can see that SVM performed the poorest when dealing with complex and high-dimensional vehicle insurance claim data, achieving an accuracy of only 54.1% and a Type II error rate of 45.5%. Naive Bayes, Logistic Regression, and BP Neural Network showed improved performance compared to SVM, with accuracy rates of 60.9%, 68.6%, and 67.2% respectively. Random Forest and Decision Tree models performed reasonably well, with accuracy rates of 75.4% and 78.4% respectively. Evidently, the PSO-XGBoost model performs exceptionally well in identifying fraudulent automobile insurance claims.

4.4. Feature importance analysis

SHAP allows interpreting the effect of the input variables in the output, and helps policymakers to interpret machine learning models; it calculates the variables importance. In this paper, we used the SHAP tool for global model interpretation. By taking the average of the absolute SHAP values for each feature and comparing them, we can determine the importance of each feature for the overall prediction. Fig. 3 displays the feature importance ranking for predicting whether a claim belongs to automobile insurance fraud in the PSO-XGBoost model.

The variables are ranked based on their importance, with higher ranks indicating greater importance. ClaimSize is the most important variable, meaning that the amount of the claim made by the policyholder has the best global explanatory power for determining whether it is a case of automobile insurance fraud. Following that, variables such as VehicleCategory, Fault, and Age also play significant roles in the model's prediction. The importance levels of VehiclePrice and Make are similar. Variables like PoliceReportFiled and WitnessPresent are ranked lower, suggesting that the presence of a police accident report or eyewitnesses is less important for the model's prediction. However, this doesn't necessarily mean that these variables have minimal impact on the model's prediction. It indicates that compared to the previous features, particularly ClaimSize, they do not provide as much information in improving the prediction accuracy.

To gain a clearer understanding of how each variable impacts the prediction, we plotted a summary of SHAP values in Fig. 4. Each feature of every sample is represented as a point on the graph, color-coded from blue (low feature value) to red (high feature value). The features are

Table 6
Comparison of performance among different classification models

Model	Accuracy	Recall	Precision	Type II error rate	F1	Handling
SVM	54.1%	54.5%	38.1%	45.5%	0.448	Sampling
Naive Bayes	60.9%	81.1%	45.8%	18.9%	0.585	Sampling
Logistic	68.6%	40.1%	56.8%	59.9%	0.470	Weighting
BP Neural Network	67.2%	39.5%	53.6%	60.5%	0.455	Sampling
Decision Tree	75.4%	71.0%	63.8%	29.0%	0.672	Pruning
Random Forest	78.4%	68.8%	65.5%	31.2%	0.671	Sampling
XGBoost	92%	92%	93%	8.0%	0.921	Sampling
PSO-XGBoost	95%	95%	95.4%	5.0%	0.950	Sampling

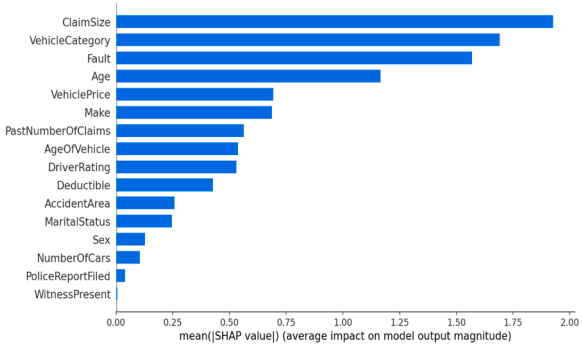


Fig. 3. Ranking of SHAP values (absolute) for all variables

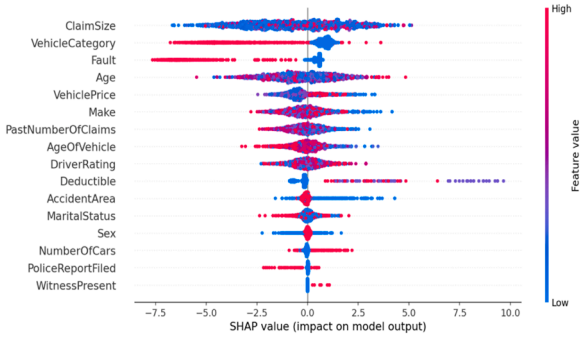


Fig. 4. Summary plot of SHAP values for the entire sample

arranged on the vertical axis in descending order of impact, while the horizontal axis measures the feature's effect on the model's output. Positive SHAP values suggest that the feature tends to increase the predicted value, whereas negative values suggest a decrease. Notably, ClaimSize has the most significant impact on predictions. The distribution of red and blue points is scattered and irregular, indicating that ClaimSize affects the model's output both positively and negatively, likely reflecting variations in the sample data itself. The second most important variable is VehicleCategory. For smaller values of VehicleCategory, there is a positive impact on the model's output, whereas larger values generally tend to have a negative impact. Nonetheless, the relationship is not completely monotonic; certain higher values can still positively affect the model's output. For instance, when the VehicleCategory is "Sedan," it shows a higher likelihood of fraud, while "Utility" indicates a lower likelihood. Regarding Fault, our study finds that when Fault is 0, indicating the policyholder's fault, it positively influences the model's output, suggesting a higher probability of insurance fraud. Conversely, when Fault is 1, indicating third-party fault, it is associated with a lower probability of insurance fraud, which aligns with our experience and common understanding.

To investigate how the variables VehicleCategory and VehiclePrice interact and affect the model's predictions, we have generated a SHAP dependency plot for the feature interaction (see Fig. 5). Fig. 5 illustrates the interaction between these two variables. The plot reveals that when the automobile price falls within the range of 20,000 to 29,000, the blue points are more concentrated, indicating a higher probability of the model predicting fraud when VehicleCategory is "Sedan." As the automobile price increases, particularly in the price range of 30,000 to 39,000, the impact becomes negative regardless of the VehicleCategory type, suggesting a decrease in the predicted probability. However, this does not imply that the impact of increasing automobile prices on the model's predictions is always negative. The plot also shows that when the automobile price exceeds 69,000 and the VehicleCategory is "Utility", the SHAP value turns positive, indicating a higher likelihood of automobile insurance fraud. The SHAP dependency plot helps to

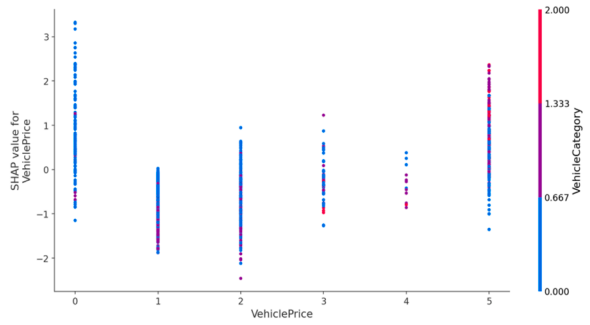


Fig. 5. SHAP dependency plot of the interaction between VehiclePrice and VehicleCategory

visualize how the interaction between VehicleCategory and VehiclePrice influences the model's predictions, providing insights into the varying effects of these variables on the predicted probabilities across different ranges.

5. Conclusion

Automobile insurance fraud is a chronic issue in the insurance sector, making effective detection strategies crucial for the industry. In this paper, we introduce an innovative approach for automobile insurance fraud detection utilizing a Particle Swarm Optimization-enhanced XGBoost model combined with interpretable machine learning method. We integrate PSO to fine-tune the parameters of the XGBoost model and systematically analyze the factors contributing to insurance fraud, aiming to improve prediction accuracy. Extensive experiments validate the effectiveness of PSO-XGBoost model in identifying fraudulent claims, demonstrating its superiority in enhancing the detection capabilities of insurance companies. By leveraging the interpretable machine learning technique, we also ensure the transparency and understandability of the fraud detection process. To better comprehend the influence of each variable on prediction outputs, we ranked their importance and depicted these using SHAP summary graphs. This paper finds that ClaimSize is the most essential factor, affecting the model's output both positively and negatively, regardless of its specific value. As for VehicleCategory, lower values positively impact the output while higher values have a negative effect, though this relationship is not entirely monotonic. Fault is another crucial element. When Fault equals 0, it increases the likelihood of insurance fraud, whereas Fault being 1 reduces this probability.

In future work, we intend to refine and adapt our model to more effectively capture complex patterns and relationships within datasets. Our goal is to reduce the model's complexity and to explore the use of automatic hyperparameter optimization techniques, thereby enhancing the model's predictive accuracy and operational efficiency.

CRediT authorship contribution statement

Ning Ding: Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Xiao Ruan:** Writing – review & editing, Visualization. **Hao Wang:** Writing – review & editing. **Yuan Liu:** Validation, Supervision, Conceptualization.

Conflict of Interest

None

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Data availability

Data will be made available on request.

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